SETTLING OPEN CASES OF THE SIGNED DIFFERENCE SET TABLE, WITH LEAN-CHECKED PROOFS: A SUMMARY

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Abstract. Of the 22,459 cases of Gordon's table of signed difference sets [1] that remain open after He and Wu [5], we settle 16,101: 16,051 have no signed difference set and 50 have one. A further 11 cases listed as open are already settled by published constructions; we list them as corrections. Every case we settle or correct is checked by the Lean 4 proof assistant. The research, the proofs and their Lean formalization were produced with AI tools under my direction; see the end of the note. This note summarizes the results, the database corrections and how to check them.

1. Results

A signed difference set (SDS) with parameters (v, k, λ) in an abelian group G of order v is a map $D\colon G \to \{0, \pm 1\}$ with k nonzero values such that $\sum_x D(x)D(x+g) = \lambda$ for every $g \neq 0$ [2]. We write $n = k - \lambda$, and $s = \pm\sum_x D(x)$ for its sign sum, whose square is $k + \lambda(v-1)$ [2, Lemma 1.1]. Most rules work with the *coset sums* $a_h = \sum_{\phi(x)=h} D(x)$ of D on a quotient $\phi\colon G \to H$, with $c = v/|H|$ (the intersection numbers of [2, Lemma 5.2]); they satisfy $|a_h| \leq c$ and $\sum_h a_h a_{h+g} = \lambda c$ for $g \neq 0$. Theorem numbers of cited papers are those of their arXiv versions, where there is one.

Table 1 lists the rules. Each is a Lean theorem that holds for every group (Z25 and Z29 each contain two rules; Z1, Z2 and Z26 settle single cases directly); a row of the table is settled by checking the rule's hypotheses for that row inside Lean. Rows are counted once, by the first rule in the order the rules were found (their numbers). "Ours" means we found no earlier statement for signed difference sets; it does not mean the idea is new.

Most rules share one shape. If an SDS existed, its coset sums on a suitable quotient H would be integers with $|a_h| \leq c$, the sum $\pm s$ and the autocorrelation above, and often further conditions: congruences from self-conjugate primes, values of characters, or invariance under multipliers. Each rule shows that for its cases no integer vector meets them all. The exceptions work on G itself: support parity (Z11) and the exhaustive searches of Z1, Z15 and Z27 (the latter two over multiplier orbits); Z33 lifts the coset sums back to G.

Table 1: Rules settling open cases (no SDS).

	Rule	Rows	Origin
Congruences from self-conjugacy			
Z8	Self-conjugacy: $p^{2j} \mid n$, $p \nmid \lvert H\rvert$, p self-conjugate modulo $\exp H$; the a_h agree modulo p^j, impossible when $p^j > 2c$	3,858	Turyn [11]; SDS form ours (odd exponent: [5])
Z9	Moments: the same congruence with $\sum a_h = \pm s$ and $\sum a_h^2 = k + \lambda(c-1)$	2,213	ours, on [2, Lemma 5.2]
Z10	Box bound: the same, with $\lvert a_h\rvert \leq c$	1,387	ours
Z22	The congruence for several self-conjugate primes at once (CRT)	132	ours
Counting and parity			

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	Rule	Rows	Origin
Z11	Support parity: v odd, λ even, $d(d-1) < v-1$ with $d = v-k$	377	classical for $\lambda = 0$ [12]; general λ ours (for $d = 1$, $v = p^m$, $p \equiv 3 \pmod 4$, m even: [5, Cor. 4.5])
Z16	Marked elements per coset (an LP certificate)	236	ours
Characters			
Z17	Real characters on a quotient C_2^r	213	extends [5, Thm. 4.2]
Z18	Characters on a quotient of exponent 4	297	ours
Z19	Characters on a quotient of exponent 3	144	classical (the ramified prime above 3); SDS form ours
Z25	Sylow quotient of order p^t: $p^{\min(t,\lceil e/2\rceil)} \mid s$, $e = v_p(n)$; for $p = 2$ a sharper equality case	222	ours, on Turyn's argument
Ma's lemma			
Z24	Ma's lemma on a quotient $C_{p^j} \times K$, p self-conjugate on K: impossible when $p^{\lfloor e/2\rfloor} > 2c$	1,457	Ma's lemma [8]; SDS form ours
Z29	Ma's lemma at its edge ($p = 2$, $4 \mid n$, on a quotient $C_{2^j} \times K$ as in Z24): $8 \mid 2s' - (k + \lambda(2c-1))$ with $s' = \sum D = \pm s$; for $c = 1$, a halved SDS	271	halving for $c = 1$: the argument of [9, Lemma 6.2] (there for weight 36); general c ours
Z32	The halved coset sums for any c (even contraction)	7	ours, on Ma's lemma
Multipliers			
Z15	Multiplier theorem ($\gcd(v,n) = 1$, n squarefree), then an exhaustive search over multiplier orbits	2,603	McFarland, as in [2, Thm. 5.1]; fixed translate [6]
Z27	The same for every n prime to v (Lean proof through Witt vectors)	1,620	as Z15
Z28	Multiplier-fixed coset sums on a quotient, searched exhaustively	716	the search of [2, §5]; its Lean proof ours
Z31	Self-conjugate primes enlarge the multiplier group of the coset sums	26	for difference sets [3, 7]; SDS form ours
Z30	Field descent without ramification: $\varphi(m)\, n \le mFc^2$ on a quotient $\mathbb{Z}_m$, $\gcd(m,n) = 1$, $F \mid m$, and every $t \equiv 1 \pmod F$ is prime to m and congruent modulo m to a power of each prime dividing n	259	Schmidt's field descent [10]; this form ours
Exhaustive search			
Z1, Z2, Z26	Exhaustive search and contraction for single small cases	4	ours
Z33	Contraction to a quotient, then a lift coset by coset	9	as in [2, §5]; the rows and the Lean proof ours

The 50 cases with a set are listed in Appendix A; Lean checks each set. The appendix's last column is the Solution's ID in `solutions.md`; each set is there as data, in `lean/SignedDifferenceSets/Answer/N<case>.lean` and as a text file listed in `objects/_objects.md` (paths relative to the problem's folder `problems/signed-difference-sets/`). They come from exhaustive and orbit searches, from cyclotomic classes, from partial difference sets ($P - \{0\}$ is an SDS when $\lambda' - \mu = 2$, the step in the proof of [5, Thm. 5.3]), and from signed combinations of relative difference sets (Bose's affine and Pott's semiregular ones, see [8]). One such combination gives, on paper, a cyclic family $((q-1)(q^2+1),\ q^2+q-1,\ q-1)$ for every prime power q; Lean checks its two members that were

open, $q = 7$ and $q = 8$. The $(27, 10, 1)$ set in C_3^3 is the construction of [4, Thm. 4.1] at $m = 1$, outside its stated range.

For example, the $(25, 12, 1)$ set in $C_5 \times C_5$ (case N49), with $D(a, b)$ in row a and column b, $a, b = 0, \ldots, 4$, and blanks for 0:

+				
				+
	+		−	−
+	+		+	
	+	−	+	+

2. Corrections to the database

These 11 cases are listed as open in the database [1], although published constructions settle them. For each we give a set checked in Lean. We found them while checking published constructions against the open cases; the list is not complete (Section 4).

Cases	Parameters	Construction
N10864, N12056, N13963, N16190, N18527, N19488, N49190	$(211, 106, 51)$, $(223, 112, 54)$, $(239, 120, 58)$, $(251, 126, 61)$, $(263, 132, 64)$, $(271, 136, 66)$, $(419, 210, 103)$, all cyclic	[2, Prop. 2.3] (Paley and zero)
N67	$(27, 14, 5)$ in C_3^3	[2, Prop. 2.3] in GF(27)
N73	$(27, 17, 8)$ in C_3^3	[2, Thm. 4.1], $q = 3$
N49624	$(421, 316, 25)$ cyclic	[4, Prop. 5.5(2)]
N14471	$(243, 82, 1)$ in C_3^5	[4, Thm. 4.1] with $x_1 = 0$

For N14471, Theorem 4.1 of [4] needs $x_1 = 0$: for $x_1 \neq 0$, as in their Example 4.2 ($x_1 = (1, 0)$), the construction is not a signed set.

3. How to check it

The Lean library lives in the repository `https://github.com/qedavid/solverz`, in the folder `problems/signed-difference-sets/lean` (Lean 4 v4.35.0 with Mathlib). The statement every result refers to is:

```
def IsSDS (k : ℕ) (lam : ℤ) (D : G → SignType) : Prop :=
  (Finset.univ.filter fun x => D x ≠ 0).card = k ∧
    ∀ g : G, g ≠ 0 → ∑ x, ((D x : ℤ) * (D (x + g) : ℤ)) = lam

def HasSDS (G : Type*) [AddCommGroup G] [Fintype G] (k : ℕ) (lam : ℤ) : Prop :=
  ∃ D : G → SignType, IsSDS k lam D
```

A single case then reads, for example (`SignedDifferenceSets.Answer.N3.not_hasSDS`),

```
theorem not_hasSDS : ¬ HasSDS (ZMod 3 × ZMod 3) 8 1
```

Every case's statement comes from its table entry, with the group $\mathbb{Z}_{m_1} \times \cdots \times \mathbb{Z}_{m_r}$ of its invariant factors $[m_1, \ldots, m_r]$: written out, as above, or through one definition `Grp` that unfolds to that product (Lean proves it has $m_1 \cdots m_r$ elements). So besides the definition of an SDS, a reader needs to check only that each statement's group, k and λ match its table entry, which a script can compare.

From the repository root,

```
lake build --wfail SignedDifferenceSets
```

checks every result. For any result, `#print axioms` shows only the three standard axioms `propext`, `Classical.choice` and `Quot.sound`. No result uses `native_decide`: every computation is checked by Lean's kernel. A build from scratch takes hours, and single modules need up to 12.3 GB of RAM; the problem's README says how to build it on smaller machines.

4. What is still open

6,347 cases remain open in the database after our results: 19 with $v < 50$, 194 with $50 \le v < 100$, 790 with $100 \le v < 200$ and 5,344 with $200 \le v < 500$. Roughly, they fall into four kinds: cases where a multiplier applies but the orbit search is too large; cases where a rule above fails by a small margin; p-groups with n a power of p, where a construction seems more likely; and a core where no multiplier applies and every congruence modulus is below c. We expect most of the core to have no SDS, but none of the local arguments above can show it. Work on these cases continues; results beyond this summary, including further corrections to the database, will be reported separately.

Use of AI

I did this work with AI tools (Anthropic's Claude models). They searched the literature, found and proved the nonexistence rules, ran the searches, wrote the Lean proofs and drafted this note; I directed the work and decided which results to keep. So that nothing rests on the AI alone, every case we settle or correct is checked by Lean 4, with no unproved steps and no axioms beyond the three standard ones (the cyclic family above and the remark on [4, Thm. 4.1] are on paper), and before I accepted a result a separate AI agent tried to break it: its proof, its Lean statement, the cases it settles and its credit to earlier work; these reports are public too, in the problem's folder `reviews/`. The Lean statements are the authoritative form of every case we settle or correct; all code, proofs and data are public (`https://github.com/qedavid/solverz`).

Appendix A. The cases settled with a set

Case	(v, k, λ)	Group	Solution
N49	$(25, 12, 1)$	C_5^2	Z3
N61	$(27, 10, 1)$	C_3^3	Z4
N62	$(27, 10, 1)$	$C_3 \times C_9$	Z5
N68	$(27, 14, 5)$	$C_3 \times C_9$	Z6
N127	$(32, 20, 4)$	C_2^5	Z12
N128	$(32, 20, 4)$	$C_2^3 \times C_4$	Z12
N129	$(32, 20, 4)$	$C_2^2 \times C_8$	Z12
N130	$(32, 20, 4)$	$C_2 \times C_4^2$	Z12
N131	$(32, 20, 4)$	$C_2 \times C_{16}$	Z12
N132	$(32, 20, 4)$	$C_4 \times C_8$	Z12
N137	$(32, 28, 12)$	$C_2 \times C_4^2$	Z12
N139	$(32, 28, 12)$	$C_4 \times C_8$	Z12
N169	$(36, 11, 2)$	$C_3 \times C_{12}$	Z12
N170	$(36, 11, 2)$	C_6^2	Z12
N286	$(45, 37, 1)$	$C_3 \times C_{15}$	Z13
N352	$(49, 33, 1)$	C_7^2	Z13
N638	$(63, 47, 11)$	$C_3 \times C_{21}$	Z13
N658	$(64, 18, 2)$	C_2^6	Z23
N659	$(64, 18, 2)$	$C_2^4 \times C_4$	Z23
N660	$(64, 18, 2)$	$C_2^3 \times C_8$	Z23
N661	$(64, 18, 2)$	$C_2^2 \times C_4^2$	Z23
N663	$(64, 18, 2)$	$C_2 \times C_4 \times C_8$	Z23
N665	$(64, 18, 2)$	C_4^3	Z23
N672	$(64, 22, 6)$	$C_2^2 \times C_4^2$	Z14
N678	$(64, 22, 6)$	C_8^2	Z14
N735	$(64, 46, 30)$	C_2^6	Z14
N736	$(64, 46, 30)$	$C_2^4 \times C_4$	Z23
N737	$(64, 46, 30)$	$C_2^3 \times C_8$	Z23

Case	(v, k, λ)	Group	Solution
N738	$(64, 46, 30)$	$C_2^2 \times C_4^2$	Z14
N740	$(64, 46, 30)$	$C_2 \times C_4 \times C_8$	Z23
N742	$(64, 46, 30)$	C_4^3	Z23
N744	$(64, 46, 30)$	C_8^2	Z14
N1164	$(79, 61, 38)$	C_{79}	Z20
N3243	$(125, 28, 3)$	$C_5 \times C_{25}$	Z23
N3314	$(125, 97, 72)$	C_5^3	Z23
N3315	$(125, 97, 72)$	$C_5 \times C_{25}$	Z23
N3486	$(127, 109, 90)$	C_{127}	Z20
N11650	$(217, 193, 168)$	C_{217}	Z20
N11991	$(222, 69, 20)$	C_{222}	Z20
N16957	$(256, 106, 42)$	C_{16}^2	Z14
N17201	$(256, 154, 90)$	C_2^8	Z14
N24177	$(300, 55, 6)$	C_{300}	Z21
N26075	$(312, 300, 200)$	C_{312}	Z23
N31721	$(343, 54, 5)$	C_7^3	Z23
N31722	$(343, 54, 5)$	$C_7 \times C_{49}$	Z23
N32087	$(343, 289, 240)$	C_7^3	Z23
N32088	$(343, 289, 240)$	$C_7 \times C_{49}$	Z23
N35840	$(361, 316, 141)$	C_{19}^2	Z7
N47314	$(410, 329, 248)$	C_{410}	Z23
N58470	$(455, 71, 7)$	C_{455}	Z20

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